CohortCert: Technical Companion
Full Definitions and Reproduction Parameters

Zida Yang, Zisheng Kuang, Guquan You, Yongjun Dai,
Junqin Chen, Haobing Liu, Jiexin Zheng, Cong Wang, Xiaolong Shi
First four authors share first authorship.
Corresponding author: Xiaolong Shi.

This is a separate online supplement to *CohortCert: Shared Evidence and Proof Costs for Cohort Decisions*. It is not an appendix to the conference-submission PDF. It retains the full formal definitions, validity calculations and frozen experiment parameters underlying the more accessible main exposition. It introduces no new experiment, retuning or result. The linked public repository is `https://github.com/yzd2002/CohortCert`. The main paper explains the running example, execution contract, completion program and findings without requiring this document.

Sections TC1–TC2 retain the previous complete method definitions, including the exact likelihood mixture, logical constraints, rational lower-bound check, forecast implementation and scoped mathematical diagnostics. Section TC3 records experimental parameters and comparator scores; TC4 retains the full closure/verification table. Equation and section references below are local to this companion. Numeric results and original code/data archives are unchanged.

Contents

TC1 Decision Contracts and Valid Shared Evidence . . . 2
TC1.1 Budget and Cost . . . 2
TC1.2 One Frozen Inference Kernel for All Policies . . . 3
TC2 Planning the Shared Certificate Frontier . . . 4
TC2.1 Nested Prefixes Give Additive Covering Gains . . . 4
TC2.2 A Compact Completion Program . . . 4
TC2.3 An Auditable Offline Cost Oracle . . . 5
TC2.4 Isolating Overlap Endpoint Dependence . . . 6
TC2.5 A Bounded Online Forecast Planner . . . 6
TC2.6 A Constructed Check on the Frontier's Scope . . . 8
TC2.7 Checking the Without-Replacement Assumption . . . 8
TC3 Experimental Design . . . 8
TC3.1 Natural Overlap and Reference Predicates . . . 8
TC3.2 GoEmotions Source Validation . . . 9
TC3.3 Common Execution and Comparators . . . 10
TC3.4 A Fixed Released-Judge Vector Request . . . 11
TC3.5 Metrics, Audits and Uncertainty . . . 12
TC3.6 Oracle, Replications and Runtime Scope . . . 13
TC4 Complete Closure and Verification Flags . . . 13

TC1 Decision Contracts and Valid Shared Evidence

A request specifies a finite text snapshot, one versioned binary label operator y_i, metadata-defined cohorts G_g with known positive denominators n_g, decision thresholds, a call budget B and risk α. It returns a justified value or unresolved marker for each item, the selected set for a completed ranking, and its acquisition cost. A cohort rate is $\mu_g = \sum_i [i \in G_g] y_i / n_g$. The finite label vector is the inference target; changing crowd to model labels changes that target. Cohort overlap is represented exactly: in $\mu_A - \mu_B$, a shared record has weight $1/n_A - 1/n_B$, which cancels only when the denominators agree.

The compiler produces rational contrasts $\theta = Cy$ and the effective support $I = \{i : \exists j, C_{ji} \neq 0\}$. After observing a pilot P, it groups residual records with identical coefficient columns into cells. Cell h has size N_h and unknown positive count K_h. Each declared decision has a positive witness family

$$D_d = \bigvee_{w \in \mathcal{W}_d} \bigwedge_{j \in J_w} \mathcal{A}_j, \qquad \mathcal{A}_j : \ \beta_j + \sum_h a_{jh} K_h > 0. \tag{1}$$

For a lower comparison $\theta_j > b_j$, choose a positive common-denominator multiplier γ_j, set $a_{jh} = \gamma_j C_{ji}$ for i in cell h, and set $\beta_j = \gamma_j(\sum_{i \in P} C_{ji} y_i - b_j)$. Cohort denominators stay fixed; observed pilot contributions enter β_j. Upper comparisons reverse signs. Equivalence with tolerance $\tau > 0$ is $(\theta + \tau > 0) \wedge (\tau - \theta > 0)$, so boundary equality is false. All atom coefficients and propagated endpoints are integers. Certification is then exact on this lattice: atom j is true if $L_j \geq 1$, false if $U_j \leq 0$, and otherwise unresolved. Equality belongs to the false side of a strict atom.

A true decision needs one witness whose atoms are all certified true. A false decision needs a certified false atom in *every* positive witness:

$$\neg D_d = \bigwedge_{w \in \mathcal{W}_d} \bigvee_{j \in J_w} \neg \mathcal{A}_j. \tag{2}$$

This cover is stored directly; expanding its negative disjunctive normal form is unnecessary. Every item needs its own justified value. For strict top-k, each k-element set S is a witness with atoms $\mu_a - \mu_b > 0$ for all $a \in S, b \notin S$. A completed request returns that S; only positive identification is allowed, including inside mixed requests. No ordering within S is required. Ties may leave strict-set identification unresolved even at census.

A *decision certificate* consists of these asserted atom sides and their count-interval support. This contract uses *atomwise interval proofs*. For example, $\theta > 0$ and $\theta < 0$ are jointly incompatible, but an interval spanning zero refutes neither atom individually. Joint logical reasoning can therefore certify more than our atomwise rules. Our optimization results concern this stated proof system.

TC1.1 Budget and Cost

The request fixes a semantic-call cap B and risk α before execution. A call is a submitted record–predicate label attempt, including pilots and failed batches. The objective is correct complete output within B and request-level risk α; unresolved requests stay in the denominator. Within one primary request, each record–predicate label is acquired once and reused by every relevant contrast. Other requests/predicates are charged afresh. Controls share this evidence structure and support filtering. Cohort metadata are given; feature preparation and physical forwards/tokens/time are accounted separately.

TC1.2 One Frozen Inference Kernel for All Policies

A charged pilot is sampled uniformly from the support. Its exact contrast contribution is removed, and the residual records are partitioned by identical coefficient columns. The primary revision uses one cell per coefficient signature. A pilot-fitted text proxy fixes a working predictive law; a constant smoothed pilot probability handles a single-class pilot. No unrevealed labels enter the proxy fit. All cells, priors, risks and formal permutations are then frozen.

Let K_h be the unknown positive count of a residual cell of size N_h. A fresh independent uniform permutation supplies each cell's formal observations. After n reads containing s positives, the ordered-path likelihood under candidate K is

$$p_{h,K}(y_{1:n}) = \frac{(K)_s(N_h - K)_{n-s}}{(N_h)_n}. \tag{3}$$

Here $(x)_r = x(x-1)\cdots(x-r+1)$, with $(x)_0 = 1$; the ordered-path probability has no binomial factor. Following sampling-without-replacement confidence sequences [1], invert $q_h(y_{1:n})/p_{h,K}(y_{1:n}) < 1/\alpha_h$, where q_h is a fixed-prior mixture of uniform-Beta and pilot-proxy Beta path laws. With Euler beta function B, this path distribution is

$$q_h = (1-\lambda)\frac{\mathrm{B}(1+s, 1+n-s)}{\mathrm{B}(1,1)} + \lambda\frac{\mathrm{B}(a_h+s, b_h+n-s)}{\mathrm{B}(a_h, b_h)},$$

where $\lambda = 1/2$, the proxy adds 20 pseudo-observations to Beta$(1,1)$ (total concentration 22), and a_h, b_h are frozen pilot-based shapes. Component *prior* weights are fixed; conditional predictive weights update by Bayes' rule. The per-cell risk is uniformly allocated with downward numerical rounding so $\sum_h \alpha_h \le \alpha$.

For the predictable next-label null law p_K and conditional predictive alternative q_t, null-possible histories satisfy

$$\mathbb{E}_K\left[\frac{q_t(Y_t \mid Y_{<t})}{p_K(Y_t \mid Y_{<t})}\,\middle|\, Y_{<t}\right] = \sum_{z:p_K(z|Y_{<t})>0} q_t(z \mid Y_{<t}) \le 1. \tag{4}$$

Hence $q_h/p_{h,K}$ is a nonnegative supermartingale even with shrinking null support. Null-impossible prefixes are excluded directly; Ville inversion applies.

At each prefix, accept the integers $K \in [s, s+N_h-n]$ satisfying the ratio inequality. Convex log ratios give a contiguous accepted interval. Starting at $[0, N_h]$, the executor intersects its endpoints with the preceding interval after *every record*, including inside a submitted batch. An empty intersection is a sticky failure: later reads cannot erase it. At exhaustion the exact count is intersected with the preceding interval, rather than replacing an earlier exclusion. The resulting $[\ell_h(n), u_h(n)]$ depends only on that cell's revealed prefix and frozen parameters. For each atom, exact signed propagation gives

$$L_j = \beta_j + \sum_{a_{jh}\ge 0} a_{jh}\ell_h + \sum_{a_{jh}<0} a_{jh}u_h, \tag{5}$$

$$U_j = \beta_j + \sum_{a_{jh}\ge 0} a_{jh}u_h + \sum_{a_{jh}<0} a_{jh}\ell_h. \tag{6}$$

Condition on the snapshot, fixed labels, pilot and frozen partition. Under independent uniform formal permutations and exact likelihood inversion, Ville's inequality and a cell-wise union bound give

$$\Pr\{\forall h, n:\ K_h \in [\ell_h(n), u_h(n)]\} \ge 1-\alpha. \tag{7}$$

On this event all returned atom assertions and decision values are correct. Atoms and witnesses reuse covered counts, requiring no extra multiplicity charge. Coverage is per request, not workload-wide. An adaptive policy can change which cell advances, but receives neither future labels nor unread permutation information. This uses established inference; numerical implementation is audited separately from the exact-arithmetic guarantee.

TC2 Planning the Shared Certificate Frontier

TC2.1 Nested Prefixes Give Additive Covering Gains

Fix a current nonfailed state and a nested continuation for every cell. Locally write $\ell_h(m) = \ell_h(n_h + m)$ and $u_h(m) = u_h(n_h + m)$, so zero denotes the current state and $0 \le m \le N_h - n_h$. For m additional reads, define

$$g^+_{jh}(m) = \begin{cases} a_{jh}(\ell_h(m) - \ell_h(0)), & a_{jh} \ge 0, \\ -a_{jh}(u_h(0) - u_h(m)), & a_{jh} < 0, \end{cases} \tag{8}$$

$$g^-_{jh}(m) = \begin{cases} a_{jh}(u_h(0) - u_h(m)), & a_{jh} \ge 0, \\ -a_{jh}(\ell_h(m) - \ell_h(0)), & a_{jh} < 0. \end{cases} \tag{9}$$

All gains are nonnegative and nondecreasing up to a sticky failure. Direct expansion gives

$$L_j(\mathbf{m}) = L_j(0) + \sum_h g^+_{jh}(m_h), \tag{10}$$

$$U_j(\mathbf{m}) = U_j(0) - \sum_h g^-_{jh}(m_h). \tag{11}$$

Thus true and false atom proofs respectively require gains of at least $d^+_j = \max\{1 - L_j(0), 0\}$ and $d^-_j = \max\{U_j(0), 0\}$. One cell can advance several atoms and decisions in either direction.

TC2.2 A Compact Completion Program

For cell h, option k represents m_{hk} additional reads, costs $c_{hk} = m_{hk}$ and supplies their gains. The full frontier allows every legal integer prefix; grid versions retain a subset including zero. Binary x_{hk} selects one option per cell; T_j, F_j select justified atom sides (both may be zero); z_w selects a witness; p_d, r_d choose a true/false decision proof. Unsupported sides cannot be selected. The cost

objective is $\min \sum_{h,k} c_{hk} x_{hk}$, subject to

$$\sum_k x_{hk} = 1 \qquad \forall h, \tag{12a}$$

$$\sum_{h,k} g^+_{jh}(m_{hk}) x_{hk} \geq d^+_j T_j \qquad \forall j, \tag{12b}$$

$$\sum_{h,k} g^-_{jh}(m_{hk}) x_{hk} \geq d^-_j F_j \qquad \forall j, \tag{12c}$$

$$T_j + F_j \leq 1 \qquad \forall j, \tag{12d}$$

$$z_w \leq T_j \qquad \forall w,\ j \in J_w, \tag{12e}$$

$$\sum_{w \in \mathcal{W}_d} z_w \geq p_d \qquad \forall d, \tag{12f}$$

$$\sum_{j \in J_w} F_j \geq r_d \qquad \forall d,\ w \in \mathcal{W}_d, \tag{12g}$$

$$p_d + r_d = 1 \qquad \forall d. \tag{12h}$$

For a strict ranking-identification decision set $r_d = 0$. With M total prefix options, J atoms, W witnesses, D decisions and E atom–witness incidences, the program uses $O(M + J + W + D)$ variables and $O(H + J + W + D + E)$ constraints. This representation avoids a negative-DNF expansion; it does not make integer optimization polynomial.

For disjoint pairs, $D = \bigvee_{w=1}^{W} (\mathcal{A}_{w1} \wedge \mathcal{A}_{w2})$ has 2^W negative-DNF terms but only W negative clauses. Primary nonranking decisions have one positive witness, and rankings forbid negative completion. Thus the experiments test shared completion/allocation; the generic representation benefit is separate.

Proposition 1 (exactness). On fixed nonfailed nested paths, a prefix allocation is feasible in (12) if and only if its intervals complete every declared decision under the stated atomwise proof system.

Proof. For a complete allocation, mark its proven atom sides, select one all-true witness for each true decision, and select the false hits for each false decision. The gain identities verify the inequalities. Conversely, the gain constraints establish each selected atom assertion, and the witness constraints implement precisely one complete proof side for each decision. Ranking masks retain identification. □

TC2.3 An Auditable Offline Cost Oracle

Only the offline evaluator sees full permutations and realized cell paths. Legal prefixes precede the first sticky inconsistency. Let C^* be the minimum additional-read cost of (12) on all legal prefixes. Every completing policy under these conditions pays at least C^*. An omniscient scheduler can attain that prefix vector in any order by nesting. Mandatory batching/exploration may add cost. Total cost adds prior submissions; our offline study starts after pilot P, giving $|P| + C^*$.

This conditional frontier fixes partition, kernel, risks and proof rules. On noncovering paths, the raw oracle may use a source-wrong atom. Fixing $F_j = 0$ for source-true atoms and $T_j = 0$ for source-false atoms defines the source-valid frontier C^*_{src}. Its selected assertions match the fixed reference, without requiring all cell intervals to cover. Hindsight witness choice is existential: online stopping may choose differently, although no such unattainability is observed on our contracts.

For general uncapped instances, Set Cover reduces to completion: a label-one unit cell represents each distinct set; each element's atom compares its containing-cell mean to an observed zero anchor. Completing all atoms selects a minimum cover. Cohort/cell counts grow in this reduction.

A feasible grid uses actual prefix costs/gains, giving an upper bound. An optimistic grid grants the next endpoint's gains at the preceding endpoint's cost; its exact integer optimum is a lower bound. Rounding that optimum up adds at most its selected bracket widths. LP-based total gaps also include integrality and solver gaps.

We verify lower bounds independently of floating solver claims. For the chosen raw or restricted formulation, let x concatenate prefix and auxiliary variables, encode non-onehot equalities in both inequality directions, and write all such constraints and fixes as $Gx \geq b$. Let $\bar{c} = c - G^\top \eta$. For any rational $\eta \geq 0$, exact weak duality gives

$$\mathcal{L}(\eta) = \eta^\top b + \sum_h \min_{k \in h} \bar{c}_k + \sum_{j \in \text{aux}} \min\{0, \bar{c}_j\} \leq \text{OPT}. \tag{13}$$

This minimizes the Lagrangian over each onehot group and auxiliary $[0, 1]$ domain. LP multipliers are converted to nonnegative rationals and checked exactly; an integer-verified allocation supplies the upper bound. These cost-bound certificates differ from online decision certificates. Wide/unavailable bounds remain in the results.

TC2.4 Isolating Overlap Endpoint Dependence

Keep the count intervals and zero-net support filtering fixed, but propagate a nonzero intersection contribution through separate left/right endpoints. For one unscaled contrast, this relaxes its signed interval $[L, U]$ to $[L - P, U + P]$, where

$$P = \sum_{h:\, h \subseteq G_A \cap G_B,\ a_h \neq 0} \min\{1/n_A, 1/n_B\}\,(u_h - \ell_h). \tag{14}$$

This identity follows by subtracting the two endpoint expressions. The relaxation leaves the pilot contribution exact. It preserves coverage but may delay either proof direction. Every source-valid relaxed proof is also a signed proof, hence $C^*_{\text{src,signed}} \leq C^*_{\text{src,relaxed}}$. Writing $\underline{C}, \overline{C}$ for total-cost bounds, a checked inequality $\overline{C}_{\text{signed}} < \underline{C}_{\text{relaxed}}$ certifies a positive additional proof-cost barrier, rather than estimating it from point bounds.

The same physical cell retains one paid prefix through two linked appearances; zero-net contributions stay zero. Only endpoint propagation changes, isolating signed overlap rather than all acquisition reuse.

TC2.5 A Bounded Online Forecast Planner

Forecast maps current evidence to a hypothetical completion allocation. Define count width $v_h(n) = u_h(n) - \ell_h(n)$ and center $s + (N_h - n)\mathbb{E}_q[p \mid y_{1:n}]$, using the mixture's posterior component weights. Round this total half-up and clip it to the current interval. Prefix zero keeps that interval; for $0 < n < N_h$ and $0 < m \leq N_h - n$, predicted width is

$$\widetilde{v}_h(n + m) = v_h(n)\sqrt{\frac{n(N_h - n - m)}{(n + m)(N_h - n)}}. \tag{15}$$

Inward integer rounding and sequential intersection make forecasts nested; exhaustion is the center singleton, and $n = 0$ uses $v_h(0)(N_h - m)/N_h$. These are scheduling predictions, never inference bounds.

Start from every cell's longest forecast prefix; if incomplete, there is no feasible forecast plan. Otherwise visit cells by decreasing longest-prefix cost, then cell ID, and binary-search the smallest option preserving complete-request feasibility. This deadline-limited coordinate pass retains a

Algorithm 1 Shared-evidence executor (all allocation policies)

Compile contrasts; charge and observe pilot P.
Freeze cells, priors, risks and independent cell permutations.
Initialize round $t = 0$ and intervals $[0, N_h]$.
loop
  if a sticky failure occurs **then**
    Return error and submitted cost.
  end if
  Propagate signed intervals; check all decision certificates.
  if every decision is justified **then**
    Return values/sets, proofs and cost.
  end if
  if cap spent or no unread cell remains **then**
    Return unresolved items and cost.
  end if
  $t \leftarrow t + 1$.
  if t is divisible by four **then**
    Choose uniformly among nonexhausted cells (seeded).
  else if an unvisited cell remains **then**
    Choose largest N_h, breaking ties by largest cell ID.
  else
    Ask the allocator for a cell and batch size.
  end if
  Use 8 reads for a first visit, otherwise at most 16; Forecast may request fewer. Clip to unread count and remaining cap.
  Charge the full batch; acquire its next formal prefix.
  Update count intervals after each returned record.
end loop

checked feasible incumbent; it is coordinatewise minimal only when the pass finishes. The compact solver may replace it only with a cheaper, exactly checked forecast allocation. Actual decisions still require observed intervals. A retained allocation schedules the cell maximizing its unfulfilled planned reads divided by remaining population, then unfulfilled reads, then smallest cell ID. Its next batch is clipped to that allocation. Between replans it keeps this allocation; fallback uses it when available, otherwise DifficultyGreedy. The clipped/deduplicated grid is $\{0, 16, 32, 64, 128, N_h - n_h\}$, with a 50 ms solver limit, 150 ms soft planning limit, one thread and replanning after eight rounds or when no planned reads remain. This period was chosen among $1, 4, 8$ on four earlier airline development queries. Soft deadlines can change rerun actions; these are scheduling settings, not efficiency guarantees.

Algorithm 1 gives the common execution loop. Policies see observed states and frozen parameters; unread labels, identities and permutations are hidden. A forecast may exceed the remaining budget, but each submitted action is clipped to the cap. Completion is checked after a successful batch: all its labels are charged and processed even if an intermediate prefix would suffice. Failed/partial batches retain all charges, return error and suppress claims. A sticky inconsistency anywhere in that batch suppresses output. Rendering failures also preserve charges.

TC2.6 A Constructed Check on the Frontier's Scope

Consider a hidden all-zero population: three disjoint 100-record cohorts A, B, X, fourteen flags per record, and 28 strict comparisons $\mu_{i,m} > \mu_{X,m}$ $(i = A, B)$. One read reveals a whole vector. Uniform five-record pilots cost 15, leaving 95 per cohort; subsequent within-cohort draws are uniform. This mathematical common-right-anchor request is separate from MAST.

Table 1: Separate all-zero construction: total vector acquisitions, including 15 pilot records. All rows use global risk 0.05. Only the first two compare the same strict request; the last changes both threshold and predictor. These are mathematical costs, not measured MAST results.

Proof family	Request	Acquisitions
Specified 42-stream family	$\theta > 0$	205 (exact)
OR count certificate	Same $\theta > 0$	≤ 197
Beta$(1, 1)$ count intervals	Changed: $\theta > 0.05$	≤ 171

The specified family retains history-feasible counts with each marginal wealth $e_h = q_h / p_{h,K_h} < 840$ and $\sum_h e_h < 840$, plus sound affine consequences ($42 = 3 \times 14$ streams). An unseen A/B positive stays feasible since $95 + 41 < 840$; both cohorts must be exhausted, costing 205.

A different OR-of-flags certificate uses $e_i(n) = \mathbf{1}\{\text{no positive OR seen}\}\, 95/(95 - n)$, $n < 95$, after zero pilots. Under the union positive-count null, $\min(e_A, e_B)$ is dominated by a true-null process. Threshold 20 is crossed after 91 reads per A/B, costing 197; positives trigger census fallback subject to the cap. Zero A/B rates settle all comparisons false regardless of X.

The changed threshold 0.05 with Beta$(1, 1)$ prediction and per-stream risk $1/840$ instead needs 78 zero reads per A/B, costing 171. The artifacts give full validity/fallback derivations and finite checks. At cap 200, these cases illustrate how proof family and task tolerance change feasibility.

TC2.7 Checking the Without-Replacement Assumption

Our kernel uses the predictable next-label null law in Section TC1.2. A scoped check explains this choice. The WOR transfer displayed by Ryu and Wornell [2, Sec. 3.2, Prop. 10] substitutes the final remaining mean into historical wealth. With counts $(3, 7)$, $N = 10$, KT prior $(1/2, 1/2)$ and risk $\delta = 1/20$, two initial category-one draws occur with probability $1/15 > \delta$. Their mixture likelihood is $3/8$; substitution of remaining mean $(1/8, 7/8)$ gives wealth $(3/8)/(1/8)^2 = 24 > 20$. The predictable-null ratio is instead $(3/8)/(1/15) = 45/8$. This fixed-time exclusion diagnoses that displayed transfer, leaving WR-KT and other vector results intact. The technical artifacts retain the full diagnostic.

TC3 Experimental Design

Primary saved-label replays compare allocation; the same-path oracle and endpoint intervention measure proof opportunities and signed overlap. Earlier-corpus replays test replication, MAST compares inference architectures using vector acquisitions, and live queries measure forwards, tokens and time. The construction in Section TC2.6 is mathematical.

TC3.1 Natural Overlap and Reference Predicates

The identity-enriched CivilComments test split in WILDS [3] supplies our primary population. A fixed salted metadata hash selects 30,000 rows having at least one of eight existing mention scores

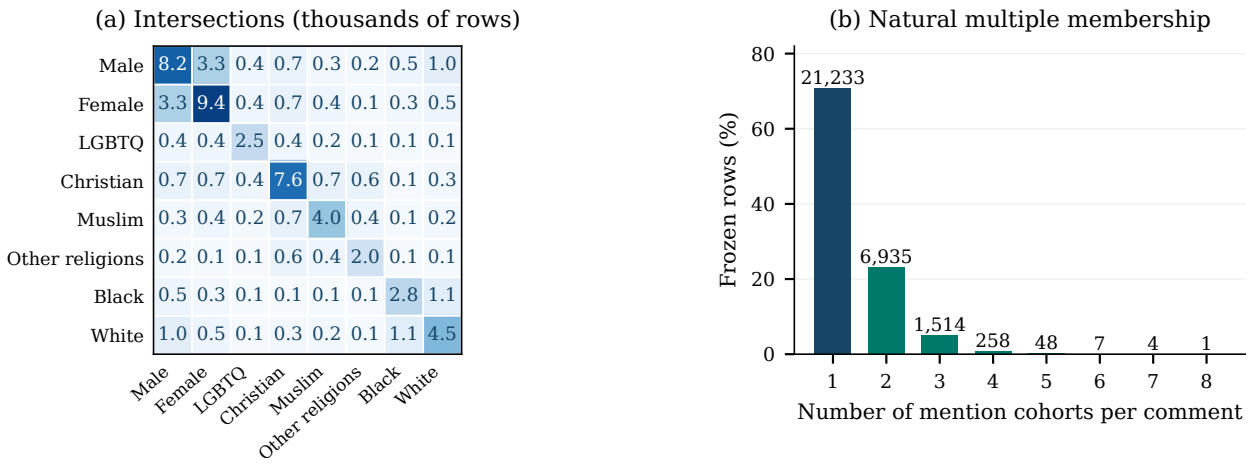


Figure 1: Metadata of the 30,000 hash-selected eligible CivilComments rows. Cohorts use existing content-mention scores ≥ 0.5. In (a), diagonal entries are cohort sizes and off-diagonal entries are intersections, rounded to thousands; exact counts accompany the artifacts. Panel (b) counts each row once. Neither panel uses target labels.

at least 0.5. Selection uses no target labels or outcomes. Duplicate texts retain distinct source IDs. Cohorts describe *content mentions*, not authors' identities or representative demographic groups. Cohort sizes are 2,011–9,409; all 28 intersections are nonempty (98–3,291 rows), and 8,767 rows (29.22%) have multiple memberships (Fig. 1). Existing toxicity, insult, identity-attack and obscene-language scores yield four crowd references, positive at decimal score ≥ 0.5, including ties.

Per predicate, the workload has 56 directed zero-margin comparisons, 28 unordered equivalence judgments (tolerance 0.05), 16 dominance summaries, 16 mixed summaries and 16 four-cohort rankings (7 top-1, 9 top-2). A dominance request asks three comparisons against one anchor; a mixed request asks one comparison and two equivalences across six distinct cohorts. Thus 116 nonranking and 16 ranking requests per predicate yield 464 primary and 64 secondary contracts. Ten seeds give 4,640 primary units and 1,280 three-decision summary units per reference/method. Truth and certifiability never filter these units.

Population, queries, thresholds, operator, original three policies and seeds were internally hash-frozen before target conversion and model inference. This is an internal freeze, not external pre-registration. Ten independent whole-workload blocks supply separate pilot, formal-permutation, exploration and planning streams, shared across methods.

A complete $30{,}000 \times 4$ Qwen3-8B matrix [4] supplies the second reference on the same population. Its pinned operator fixes weights/tokenizer, prompts, BF16, SDPA, a nonthinking chat template, one record per forward and next-token 0/1 scoring (ties positive), with no generation or KV/result reuse. All contexts fit 4,096 tokens without truncation. Operational canaries check stability separately. Allocation experiments replay this saved matrix: their charged label acquisitions are not fresh forwards. Matrix construction and live-query timing are separate studies. Both crowd and model references are fixed labels, not asserted semantic truth.

TC3.2 GoEmotions Source Validation

The three checksum-verified GoEmotions release files [5] contain 58,011 distinct IDs; 57,877 have at least two clear, distinct raters. A fixed salted metadata hash selects 30,000 eligible IDs without using emotion scores or outcomes. Four targets—anger, joy, sadness and fear—are positive when

at least half the fixed clear raters select them, including ties.

Eight metadata-only cohorts use the four most frequent subreddits within the selected 30,000 IDs, length/time at least their medians on that same selected population, and question/exclamation marks. Their sizes are $139, 125, 122, 122, 15{,}070, 15{,}000, 3{,}044, 4{,}145$. The same 132 abstract cohort-index templates transfer to each emotion, with ten fresh four-stream seeds. Inputs and policy code were hash-frozen before target conversion, after CivilComments outcomes were known. This is a separately sourced, crowd-only validation using the same request design; it introduces no new model or latency measurement.

TC3.3 Common Execution and Comparators

Here I_q denotes request q's effective support. The cap is $B = \lceil 0.5|I_q| \rceil$, request risk 0.05, and charged pilot size $\min\{256, \lfloor |I_q|/10 \rfloor, \lfloor B/4 \rfloor\}$. Label-free TF–IDF features (up to 10,000 uni/bigrams, minimum frequency two, sublinear TF) use SVD-128/L2 normalization. Pilot-only logistic regression uses $C = 1$ and at most 1,000 iterations. For an empty/single-class pilot or a fit/score failure, all residual scores become $(s_p + 0.5)/(n_p + 1)$; the fallback is recorded. Average the frozen scores within cell h to $\bar{p}_h$, and set its proxy-Beta shapes to $1 + \lfloor 20\bar{p}_h \rfloor$ and $21 - \lfloor 20\bar{p}_h \rfloor$. There is one cell per coefficient signature, with this frozen prior. All methods share it, eight-record cell initialization, later batches of sixteen, every-fourth-round exploration and the exact inference/stopping loop. Only cell allocation changes.

Matched chooses the nonexhausted cell with fewest formal records read, n_h, excluding the pilot; ties use smallest cell ID. *Forecast* uses Section TC2.5.

DifficultyGreedy prioritizes predicted progress toward an easily completed proof. It is an internal heuristic, distinct from Adaptive Dual Greedy [6]. Its defining score uses the original contrast units: let $[l_j^\theta, u_j^\theta]$ be the original contrast interval, before threshold shifting and integer scaling, and b_j its threshold. For atom j, let $W_j = u_j^\theta - l_j^\theta$, $m_j = (l_j^\theta + u_j^\theta)/2$ and $e_j = \sigma_j(m_j - b_j)$ for its original threshold b_j, where $\sigma_j = 1$ for a lower atom and -1 for an upper atom. Set

$$r_j^{\pm} = \frac{W_j}{2\max\{\pm e_j, \varepsilon\}}, \qquad \mathrm{LME}(x) = \log\left(\frac{1}{|x|}\sum_i e^{x_i}\right). \tag{16}$$

For an unresolved decision, let $\mathcal{V}_d$ be its positive witnesses not yet hit by a false atom, and R_w their unresolved atoms. Define

$$P_d = \min_{w \in \mathcal{V}_d} \mathrm{LME}(r_j^+ : j \in R_w), \tag{17}$$

$$F_d = \mathrm{LME}\left(\min_{j \in R_w} r_j^- : w \in \mathcal{V}_d\right). \tag{18}$$

The objective sums $\min(P_d, F_d)$ for binary decisions and P_d for rankings. Resolved decisions contribute zero. A blocked strict ranking with no eligible positive witness also contributes constant zero but stays incomplete. Here $\varepsilon = 10^{-4}$ and LME temperature is one; stable log-sum-exp evaluates it.

For an active cell ($n_h < N_h$), candidate batch size is $b = \min\{16, N_h - n_h, B - C_{\mathrm{spent}}\}$. Here C_{spent} is submitted calls so far. The batch predicts width reduction Δ_h using the finite-population square-root shrinkage of Section TC2.5 (no shrinkage at $n_h = 0$; zero residual width at exhaustion). Let c_{jh} be the original, unscaled contrast coefficient. Replace each W_j by $\max\{0, W_j - |c_{jh}|\Delta_h\}$, holding midpoints, unresolved atoms and witness eligibility fixed. Choose maximum predicted objective reduction divided by b; unit costs are one. Gain/cost ties within $10^{-12}\max\{1, |\mathrm{best}|\}$ use smallest cell ID. Nonpositive gain falls back to largest residual-cell size N_h, then smallest ID. The shared checker alone certifies completion.

WidthGreedy targets normalized interval reduction. It prunes settled decisions/atoms and refuted positive witnesses, then deduplicates sign/scale-equivalent live contrasts. For live directions $\mathcal{C}$, normalize $\bar{c}_{jh} = c_{jh}/R_j$ by the fixed post-pilot range $R_j = \sum_h N_h|c_{jh}|$. Its batch score is the mean predicted normalized width reduction, $|\mathcal{C}|^{-1}\sum_{j\in\mathcal{C}}|\bar{c}_{jh}|\Delta_h/b$. *VarianceGreedy* targets empirical variance reduction with the same pruning/normalization, scoring

$$\frac{1}{|\mathcal{C}|b}\sum_{j\in\mathcal{C}}\bar{c}_{jh}^2 N_h^2\widehat{S}_h^2\left(\frac{1}{n_h}-\frac{1}{n_h+b}\right), \tag{19}$$

where $\widehat{p}_h = (s_h+0.5)/(n_h+1)$ and $\widehat{S}_h^2 = N_h\widehat{p}_h(1-\widehat{p}_h)/(N_h-1)$ for $N_h > 1$. Only revealed formal counts are used. Common initialization normally supplies $n_h > 0$; otherwise relevant unobserved cells are explicitly prioritized. Both controls use relative score ties of $10^{-12}|\text{best}|$, then cell ID, and the same largest-N_h nonpositive-gain fallback. This adapts Neyman variance allocation [7].

For one unresolved scalar comparison, DifficultyGreedy's score reduces to $W/[2\max\{|m-b_0|,\varepsilon\}]$. Its predicted gain per batch cost is therefore a common positive factor times $|c_h|\Delta_h/b$. This is exactly WidthGreedy's candidate ordering in real arithmetic, even with clipped batch sizes. The threshold affects stopping but cancels from this allocation. Floating ties can differ. Single-contrast equivalence also preserves width ordering for equal-price candidates; no general unequal-price equivalence is claimed. The scalar case thus cannot establish a distinctive Boolean allocation benefit.

MaxDifficulty isolates cross-decision gain aggregation on all 128 three-decision summaries. It replaces the sum of decision-level DifficultyGreedy marginal gains by their maximum, retaining the shared evidence, checker and every other execution setting. It is not independent label acquisition. The stronger controls add 23,680 CivilComments runs: WidthGreedy/VarianceGreedy on every contract and MaxDifficulty on summaries, both sources and all seeds. These post-hoc controls were code-frozen before execution. GoEmotions uses the same controls with DifficultyGreedy: 17,120 planned runs, including summary-only MaxDifficulty; no Forecast tuning is transferred.

For CivilComments rankings, *Boundary* uses relative estimates $\widehat{\theta}_g = c_{g,o}^P + \sum_h c_{(g,o),h}N_h s_h/n_h$, with lexicographically first origin o, exact pilot difference $c_{g,o}^P$ and $\widehat{\theta}_o = 0$. It selects the top-k estimates (cohort-ID ties), then the inside cohort with smallest relative lower bound and outside cohort with largest relative upper bound, targeting their direct width reduction. It is a scoped boundary adaptation. The total is $31{,}680 + 1{,}280 + 23{,}680 + 17{,}120 = 73{,}760$ planned main-control replays; old-source, MAST and live studies are separate. Crowd/model sources are views of one population; GoEmotions is separately sourced.

TC3.4 A Fixed Released-Judge Vector Request

Each MAST-Data record [8] is a released multi-agent task trace with judge flags. Three fixed cohorts contain 100 records each: MetaGPT/GPT4o (A), MetaGPT/Claude (B), and ChatDev/GPT4o (C). Unlike the primary scalar calls, one acquisition here charges a single record and reveals its 14 released binary judge flags. Completion requires all 28 values $\mu_{A,j} > \mu_{B,j}$ and $\mu_{A,j} > \mu_{C,j}$, with ties false. Targets are released judgments, without human-gold or original-model-version claims.

Seven programs each retain eight paths. The budgets $75, 150, 225, 300$ are readouts of the same planned execution paths, rather than fresh runs at each budget. Precommitted permutations supply local pilot/residual orders, joint full-population orders and the inputs to native seeded shuffles. Prefixes across architectures may differ.

Programs. Local–RR, Width and DWidth share count confidence sequences and a complete-output checker, selecting by round robin, summed width reduction or difficulty-weighted width reduction. Five charged pilot vectors/cohort set the residual-95 mixture $\frac{1}{2}\,\mathrm{Beta}(1,1) + \frac{1}{2}\,\mathrm{Beta}(1 + s_p, 6 - s_p)$; projection retains pilot positives s_p and denominator 100. Each of 42 streams receives risk 1/840 (total 0.05).

NativeExact uses Evergreen's native seeded plans and exact cohort aggregates [9]. NativeProgressive—RR uses synchronized prefixes, full-100 inference, recorded seeds, every-prefix evaluation, grid 100, outward integer readout and sticky intersection. These are configured adapters with audited floating risks and fixed scheduling.

Joint–RR and Joint–LogGreedy share predictable without-replacement Kelly bets and verified witnesses for an entire finite maximal upper-null cover [10]. Incomplete optimization is unresolved. Classification risk is 1/560/comparison, shared by its two signs; correlated-flag capitals are not multiplied. RR cycles cohorts; LogGreedy sums predicted improvements in clipped log-evidence bottlenecks using fixed past-only Beta predictions.

Accounting and endpoint. Vector attempts, including pilots and failed/unfinished tails, are charged before return. Paid prefetch is distinguished from committed prefixes; semantic caches start empty. Credit requires resource eligibility, verified normal durable closure, exact cost $\leq B$, all 28 correct supported online outputs and no earlier source-count exclusion or erased inference error. All eight paths remain per program. Censoring means that an execution was interrupted before verified normal closure; it receives no credit. All four censors, including both Native startup censors, remain permanent. Unknown cost tails stay unknown; paths are not retried.

Exact census remains valid. PEAK/target intervals [11,12] and native repeated-draw accounting are outside this configured comparison. Vector acquisitions are not forwards, tokens or monetary costs.

TC3.5 Metrics, Audits and Uncertainty

Primary completion retains all nonranking query–seed units and requires every answer and selected proof atom to match its full reference. Correct vectors with source-invalid witnesses are separate. Wrong positive/negative claims, unresolved requests, sticky inference errors and framework errors remain. Primary costs average submitted labels/support over all planned units; a sensitivity charges errors and invalid proofs to their full caps. Jointly complete paired savings are secondary, with their selection denominator.

Balanced aggregation weights four predicates and three families equally: pairwise, equivalence and summaries, the last pooling equal-sized dominance and mixed groups. Paired 95% percentile intervals use 20,000 resamples of ten whole-workload seed blocks, conditional on source and workload. Shared records, cohorts, reversed directions and templates make run records dependent. Soft-deadline nondeterminism is not separately isolated. These intervals establish neither request-population nor corpus generalization, and do not apply to the eight-path MAST replay.

An executor/compiler-independent analyzer reconstructs fixed-reference means and Boolean families, checks proof sides/costs/fingerprints, and scans available CS paths. Preselected and outcome-selected diagnostics retain separate denominators. Source exclusion, wrong assertion, sticky failure and framework error are audited separately.

TC3.6 Oracle, Replications and Runtime Scope

The oracle selects 48 nonranking IDs by support strata/salted hashes: four predicates, comparison/equivalence/dominance/mixed subtypes and three strata. Seeds $0, 1$ and both references yield $48 \times 2 \times 2 = 192$ cases of the original three policies, with explicit source-valid atom restrictions, full-path reconstruction and matched risk/kernel. Missing/unbounded cases stay. Verified $[L, U]$ total gaps include integrality and solver gaps; floating incumbents are not lower bounds. The reported relative gap is $(\overline{C} - \underline{C})/\max\{1, \overline{C}\}$, using total costs including pilot. Later numerical bound recovery is separately versioned and applies to all 192 selected cases.

The post-hoc endpoint intervention keeps all 192 cases/realized paths. Linked virtual appearances use saved signed grids, feasible prefixes and legal endpoints. Each receives 0.25 s coordinate work, three-second LP and three-second MIP limits, one solver thread. Independent exact arithmetic checks dual lower bounds, physical single-charge costs and endpoint proofs. No online policy is rerun or retuned. For bound-bearing cases, let a_i be policy calls on a correct completion and cap B_i otherwise, and $A = \sum_i a_i$. Capped hindsight headroom is

$$\left[\frac{A - \sum_i \min(\overline{C}_i, B_i)}{A}, \quad \frac{A - \sum_i \min(\underline{C}_i, B_i)}{A}\right].$$

A proved absent source-valid proof contributes the cap at both endpoints; unavailable bounds retain separate status.

Earlier sources contain 40,000 CFPB narratives [13, 14] and 18,569 OpinRank reviews [15], four saved model predicates each. Their 480 contracts split 80/80/320 train/development/test, with disjoint test cohorts. All 320 test contracts use the revised executor, three policies, ten seeds and 50% caps (9,600 replays). These post-hoc saved-label replications have an unpinned upstream model revision and no human validation of the eight predicates. Historical one-sided results remain separate.

The metadata-hash-selected live schedule contains five CivilComments queries and one airline ranking (six cases total). The ranking uses the 14,640-row Twitter US Airline Sentiment source [16] (pinned-model negative sentiment). The warm matrix has six queries $\times$ three policies plus Boundary on two rankings (20 executions). Three fresh-process executions use the same CivilComments toxicity pairwise query, one per primary policy. Warm execution and fresh-process startup separately account physical forwards, tokens, pilot/proxy work, planning, inference, certification and rendering. Primary crowd outcomes were known; the schedule designer fixed metadata-hash selection before method-outcome inspection. Its results describe these six post-hoc cases and their warm/fresh-process conditions.

The public repository linked in the Introduction provides fixed CPU-study inputs, method and analysis sources, bounds and runtime receipts. Its manifests document external requirements, trace availability and MAST audit versions.

TC4 Complete Closure and Verification Flags

This table retains the full flag breakdown behind the simpler MAST table in the main paper. It reports the same fixed trajectories and introduces no new result. N/C count normal/interrupted closure; E marks a historical source-count exclusion and $V?$ unknown verification. The latter flags can overlap closure statuses. All eight paths remain per program. A censored path in the main table has interrupted closure and unknown verification; its missing cost is not imputed.

Table 2: Fixed 28-comparison MAST request, all original closure and audit flags. k_{300} is credited completion at budget 300; cost conditions on credited paths.

Program	$k_{300}/8$	Cost	N	C	E	$V?$
Local-CS / RR	7/8	298	7	1	0	1
Local-CS / Width	8/8	278–288	8	0	0	0
Local-CS / DWidth	8/8	278–288	8	0	0	0
NativeExact adapter	7/8	300	7	1	0	1
NativeProgressive / RR	5/8	300	7	1	2	1
Joint-Kelly / RR	8/8	298	8	0	0	0
Joint-Kelly / LogGreedy	7/8	278–290	7	1	0	1

AI-Assisted Content

The authors led the research conception, overall design and scientific writing. OpenAI Codex assisted with literature checks, code, analysis, figures, and drafting/revision of the Abstract, Sections I–VIII and Technical Companion; Qwen3-8B generated the identified model-reference labels and live-inference outputs. The authors take full responsibility for all manuscript content, including text generated or polished by LLMs, and for its accuracy, originality and compliance with research ethics, including the prevention of plagiarism and scientific misconduct.

References

[1] I. Waudby-Smith and A. Ramdas, "Confidence sequences for sampling without replacement," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 20 204–20 214. [Online]. Available: https://proceedings.neurips.cc/paper/2020/hash/e96c7de8f6390b1e6c71556e4e0a4959-Abstract.html

[2] J. J. Ryu and G. W. Wornell, "Gambling-based confidence sequences for bounded random vectors," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 42 856–42 869, also available as arXiv:2402.03683v2, revised August 21, 2024. [Online]. Available: https://proceedings.mlr.press/v235/ryu24a.html

[3] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, R. L. Phillips, I. Gao, T. Lee, E. David, I. Stavness, W. Guo, B. Earnshaw, I. Haque, S. M. Beery, J. Leskovec, A. Kundaje, E. Pierson, S. Levine, C. Finn, and P. Liang, "WILDS: A benchmark of in-the-wild distribution shifts," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 5637–5664. [Online]. Available: https://proceedings.mlr.press/v139/koh21a.html

[4] A. Yang *et al.*, "Qwen3 technical report," 2025. [Online]. Available: https://arxiv.org/abs/2505.09388

[5] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, and S. Ravi, "GoEmotions: A dataset of fine-grained emotions," in *Proceedings of the 58th Annual Meeting of*

the Association for Computational Linguistics, 2020, pp. 4040–4054. [Online]. Available: https://aclanthology.org/2020.acl-main.372/

[6] A. Deshpande, L. Hellerstein, and D. Kletenik, "Approximation algorithms for stochastic boolean function evaluation and stochastic submodular set cover," in *Proceedings of the Twenty-Fifth Annual ACM-SIAM Symposium on Discrete Algorithms*, 2014, pp. 1453–1466.

[7] J. Neyman, "On the two different aspects of the representative method: The method of stratified sampling and the method of purposive selection," *Journal of the Royal Statistical Society*, vol. 97, no. 4, pp. 558–625, 1934.

[8] M. Cemri, M. Z. Pan, S. Yang, L. A. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, and I. Stoica, "Why do multi-agent LLM systems fail?" in *The Thirty-ninth Annual Conference on Neural Information Processing Systems, Datasets and Benchmarks Track*, 2025. [Online]. Available: https://openreview.net/forum?id=fAjbYBmonr

[9] A. W. Lee, B. Han, S. Sen, S. Yeom, U. Çetintemel, and A. Datta, "Evergreen: Efficient claim verification for semantic aggregates," *Proceedings of the VLDB Endowment*, vol. 20, no. 1, pp. 16–30, 2026.

[10] J. V. Spertus, M. Sridhar, and P. B. Stark, "Sequential stratified inference for the mean," arXiv preprint arXiv:2409.06680v5, 2026. [Online]. Available: https://arxiv.org/abs/2409.06680v5

[11] B. M. Cho, K. Gan, and N. Kallus, "Peeking with PEAK: Sequential, nonparametric composite hypothesis tests for means of multiple data streams," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 8487–8509. [Online]. Available: https://proceedings.mlr.press/v235/cho24a.html

[12] S. Hait, "Target-aware sequential inference: Pooled versus stratified anytime-valid designs," 2026, arXiv:2609.37873v2. [Online]. Available: https://arxiv.org/abs/2609.37873v2

[13] Consumer Financial Protection Bureau, "Consumer complaint database," 2026, database metadata fetched 2026-10-02; see the recorded data manifest. [Online]. Available: https://www.consumerfinance.gov/data-research/consumer-complaints/

[14] ——, "CFPB consumer complaint database narratives archive," 2026, previously published narratives in the FOIA electronic reading room. [Online]. Available: https://www.consumerfinance.gov/foia-requests/foia-electronic-reading-room/cfpb-consumer-complaint-database-narratives-archive/

[15] K. Ganesan and C. Zhai. (2011) OpinRank review dataset. UCI Machine Learning Repository. Hotel subset used. DOI: 10.24432/C5QW4W. [Online]. Available: https://archive.ics.uci.edu/dataset/205/opinrank+review+dataset

[16] CrowdFlower, "Twitter US airline sentiment," Kaggle dataset, data collected in February 2015; existing crowd annotations; CC BY-NC-SA 4.0; accessed 2026. [Online]. Available: https://www.kaggle.com/datasets/crowdflower/twitter-airline-sentiment